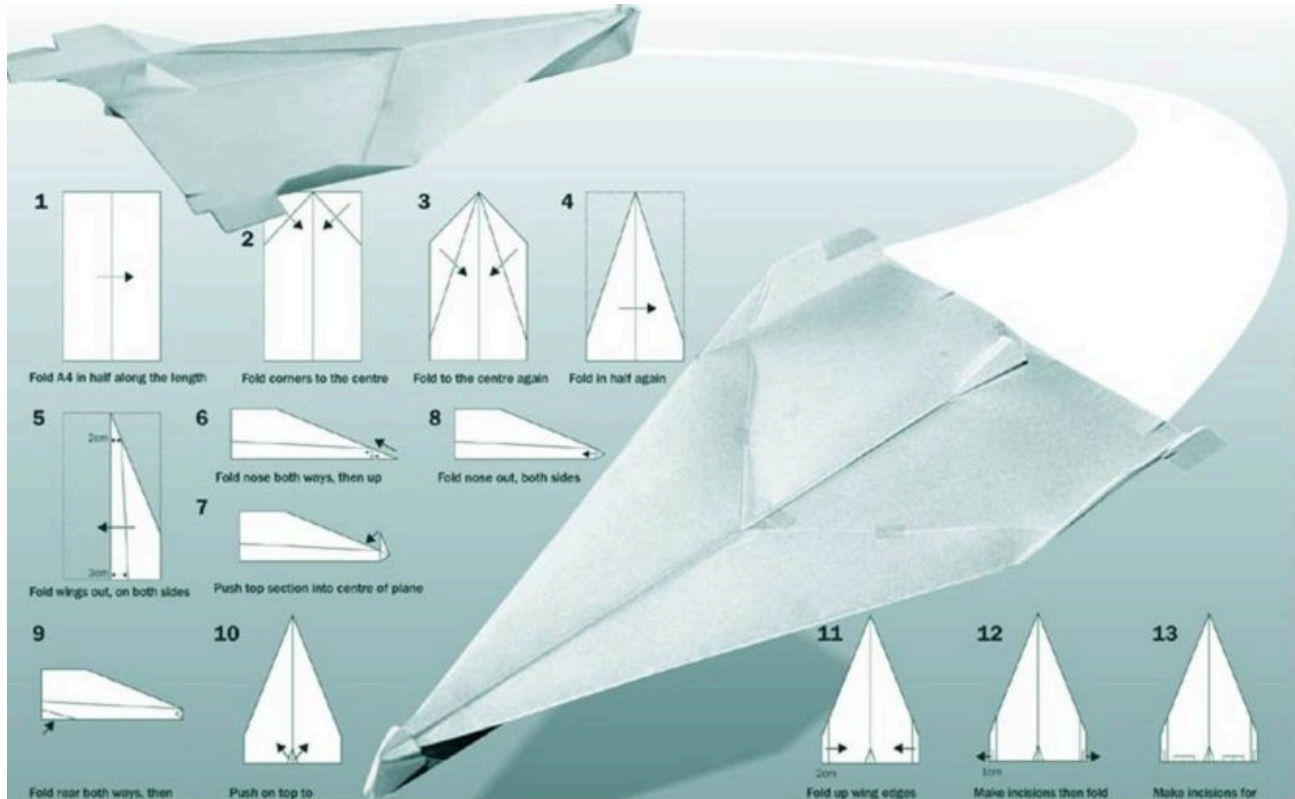


How Rocket Scientists Build Paper Airplanes – Metallicman

 metallicman.com/laoban4site/how-rocket-scientists-build-paper-airplanes

congjing yu

July 20, 2018



When I was attending university, my classmates and I would get drunk and fly paper airplanes. We were all studying engineering. Myself and my buddy Jay were both in a joint program where we studied Aerospace Engineering, while my buddy Peter studied Mechanical Engineering. All told, we had a significant amount of class overlap. As young college students, full of “piss and vinegar”, we would get into all sorts of trouble...

There in the dorms, we would launch our bottle rockets at paper airplanes that we would fly out the windows. The overall goal was to create a paper airplane that would stay aloft the longest. This was maintained by up-drafts and the design of the paper craft.

Let's look at some of the design features that we incorporated into our tiny paper devices...

Weight and Balance

One of the most important considerations for aircraft (as well as munitions) is the ability for the airplane to be well balanced. This takes into account the center of gravity of the craft relative to the uplifting force on the wing design.

Imagine walking a tight-rope. If you had a large and robust body and butt, the center of gravity would be closer to the rope. Yet if you were tall and skinny with an enormous

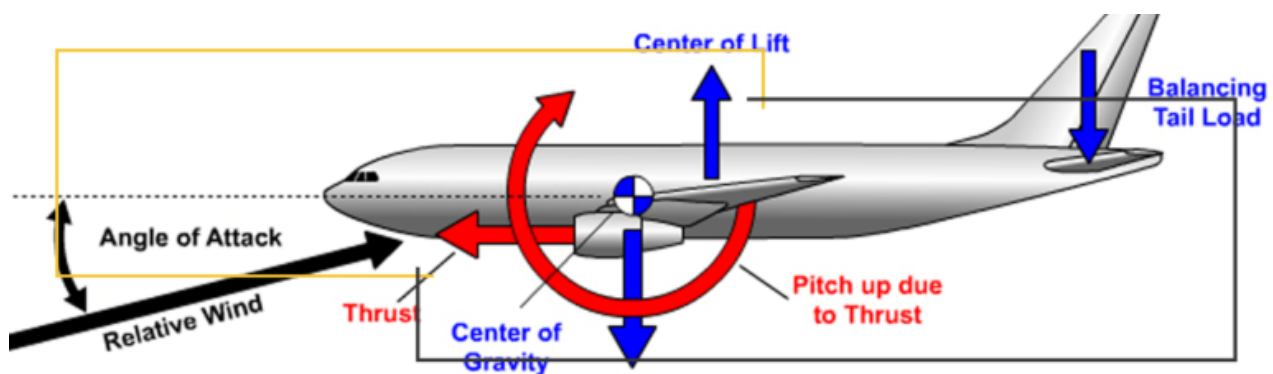
“pumpkin” head, the center of gravity would be further away from the rope. The distance between the center of gravity of the two body shapes would be different. The longer and taller person might have a more difficult time walking on the tight-rope than the fatter person.

In a like way, the center of gravity of a plane is important. There is a relationship between the center of gravity of the plane, and the lift on the wings, as well as the propulsive force at the tail.

In general placing weight forward is desirable

Placing weight forward is desirable.

The place where the paper airplane balances on the tip of your finger is called the Center of Gravity (CG). This is the point of balance. In addition, for aircraft, there is a secondary point of balance. This is a specific CG position known as the Neutral Point. This point provides neutral pitch stability.



Balancing considerations of an aircraft. Illustration shows the center of gravity relative to the center of lift.

Here is the rule of thumb. If the airplane has a CG ahead of the Neutral Point, the plane is stable, if its behind this point its unstable.

As a matter of course, all airplanes without computer assisted flight controls need a CG ahead of their neutral point. Thus almost all airplanes prior to assisted flight controls (invented in the late 1970's) were designed this way. I guess you could refer to it as a sort of “classical” aviation design.

For rectangular wings the neutral point is $\frac{1}{4}$ of the distance from the nose to the tail. For delta wings (such as the common dart paper airplane) the neutral point is $\frac{1}{2}$ of the distance from the nose to the tail.

Stability

Stability means the plane, if disturbed, will return to its original state. This is a very important behavioral feature when there are fluid dynamics that alter flight regimes. In

general, you want a stable plane. If you don't have one, you will have a crash. That would really ruin your day.

For pitch stability it means the plane will seek a single airspeed.

A plane which is unstable in pitch will either pitch up into a stall, or nose dive, but won't settle out anywhere in between. A stable airplane will tend to oscillate up and down a few times, but converge on a steady flight speed.

Many typical paper airplane designs are stable, but just... just... barely.

As a plane becomes more and more stable, it wants to fly faster and faster. To counter this tendency, up-elevator must be used to produce a good trim airspeed. This is why many of the classic paper airplane designs are nearly neutrally stable. Few people realize good pitch stability requires a heavy nose and some up-elevator.

Elevators are [flight control surfaces](#), usually at the rear of an [aircraft](#), which control the aircraft's [pitch](#), and therefore the [angle of attack](#) and the lift of the wing. The elevators are usually hinged to the [tailplane](#) or horizontal [stabilizer](#). They may be the only pitch control surface present, sometimes located at the front of the aircraft (early airplanes) or integrated into a rear "all-moving tailplane" also called a slab elevator or [stabilator](#).

-[Wikipedia](#)

The classic aircraft designs rely on the small inherent "up elevator" effect. This effect is known as a positive zero lift pitching moment. It is a direct result from the design of the wing. Often the primary factors are the swept wing, and possibly the airfoil shape.

The problem with paper airplanes is that they are initially designed without a up-elevator effect. Thus many classic paper airplanes are flown with no elevator adjustment. Sometimes they fly well, many times they don't, and they always have poor stability.

To Improve the plane you incorporate an up-elevator effect

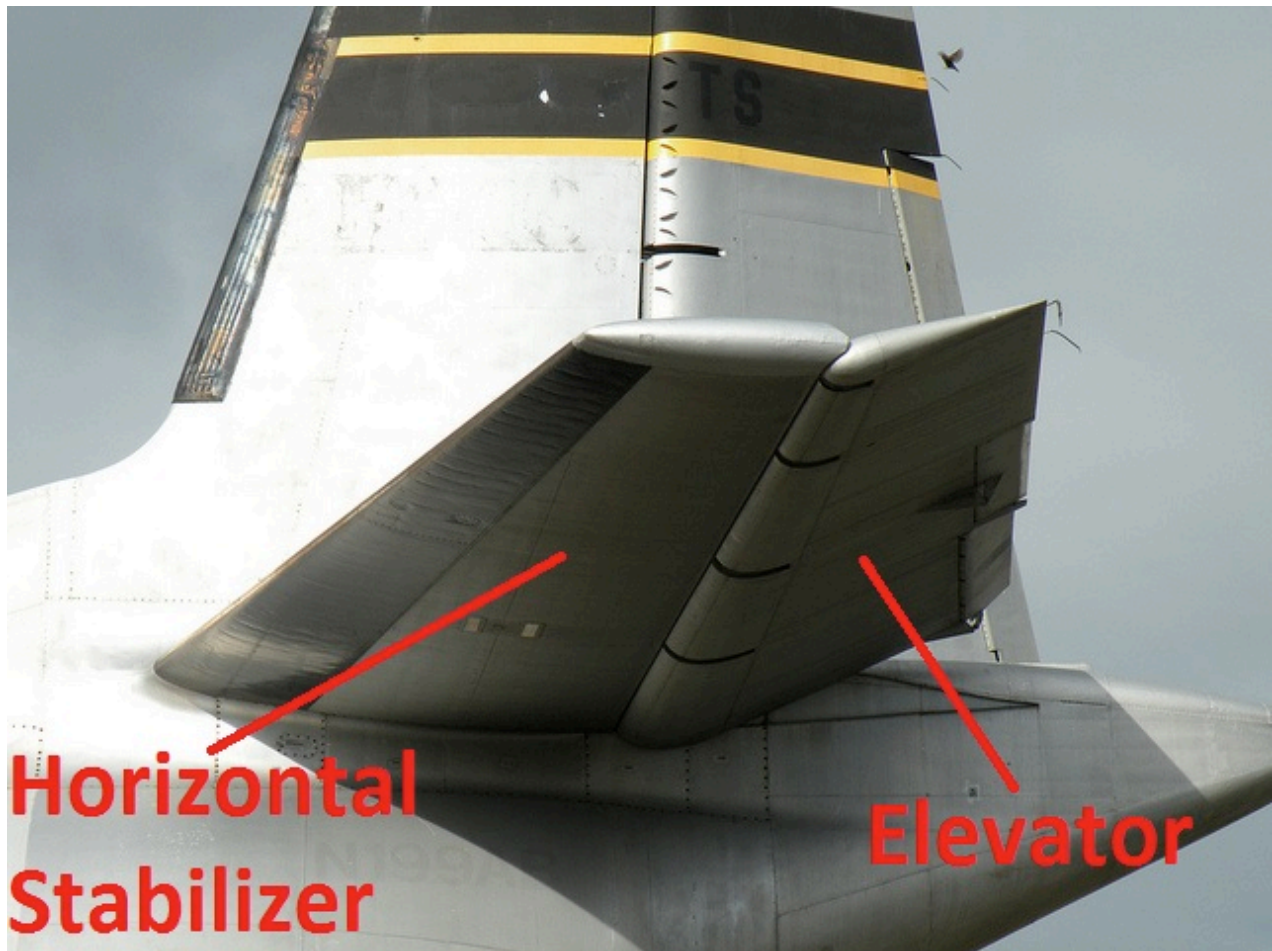
In general, the changes that one must make to a paper airplane to fly well are mostly the same adjustments which make hand launched gliders fly well.

Most people have the crazy idea that a good paper airplane needs no adjustments. You just simply fold up a piece of paper into the shape of an airfoil, and that all there is to it. You make the basic folds and then you are all finished.

This is really unfortunate.

All real airplanes have trim tabs that are used to make small adjustments to the plane. As such, it should be clear that all paper airplanes also need small adjustments to fly their

best.



The aircraft is typically “trimmed” to have it maintain balance and stability in flight. These features are called stabilizers, to well... stabilize the aircraft in flight.

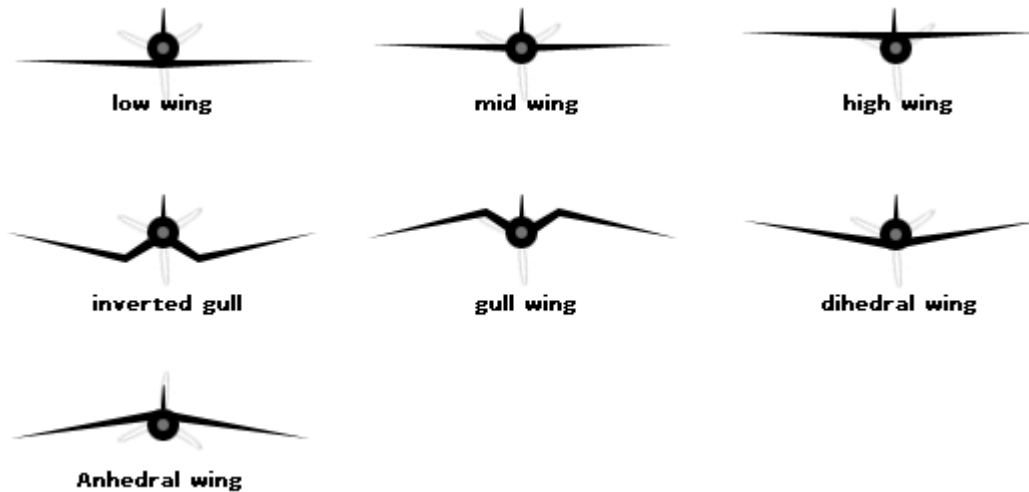
Once, the paper airplane is properly balanced, the following suggestions might help the budding paper airplane designer in keeping their work of art in the air.

Anhedral Wing Design

One of the most common paper airplane mistakes is to leave the wings folded down at an angle. Don't tell me that you don't know what I am talking about. It is perhaps the most common way of making paper airplanes.

Folding the wings down along the side at an angle is called “anhedral” shape.

The problem with an anhedral wing design is that it reduces the lateral stability of your paper airplane. To combat this, you can have a different design and shape for the wings. Instead of a anhedral wing, you can have one in a different shape.



Wing shape to include anhedral designs. For paper aircraft the wing design has a great deal of impact on the overall stability of the aircraft.

What you want is called “dihedral” wing design which is when the wing tips are the highest part of the wing. The resulting lateral stability will help keep your paper airplane flying straight. Or maybe, at the worst, in a gradual turn.

This is important, as a plane with lateral instability will tend to roll over on its back and crash. Alternatively, it might enter into an ever tightening spiral which becomes a spiraling dive. To prevent this, all you need remember is the simple rule; keep your wing tips up.

| Keep your wing tips up.

Sideslip

A **slip** is an [aerodynamic](#) state where an [aircraft](#) is moving *somewhat* sideways as well as forward relative to the oncoming airflow or [relative wind](#). In other words, for a conventional aircraft, the nose will be pointing in the opposite direction to the bank of the wing(s). The aircraft is not in [coordinated flight](#) and therefore is flying inefficiently.

-[Wikipedia](#)

Technically dihedral provides a stabilizing rolling moment due to a condition known as sideslip.

It works like this, if the plane yaws to the left (positive sideslip), the right wing has a slightly increased angle of attack (AOA) because of the dihedral. As this is going on, the left wing's AOA is decreased (this is most easily imagined if you think about 90 degrees of sideslip). As a result, the resulting rolling moment is to the left, which is stabilizing.

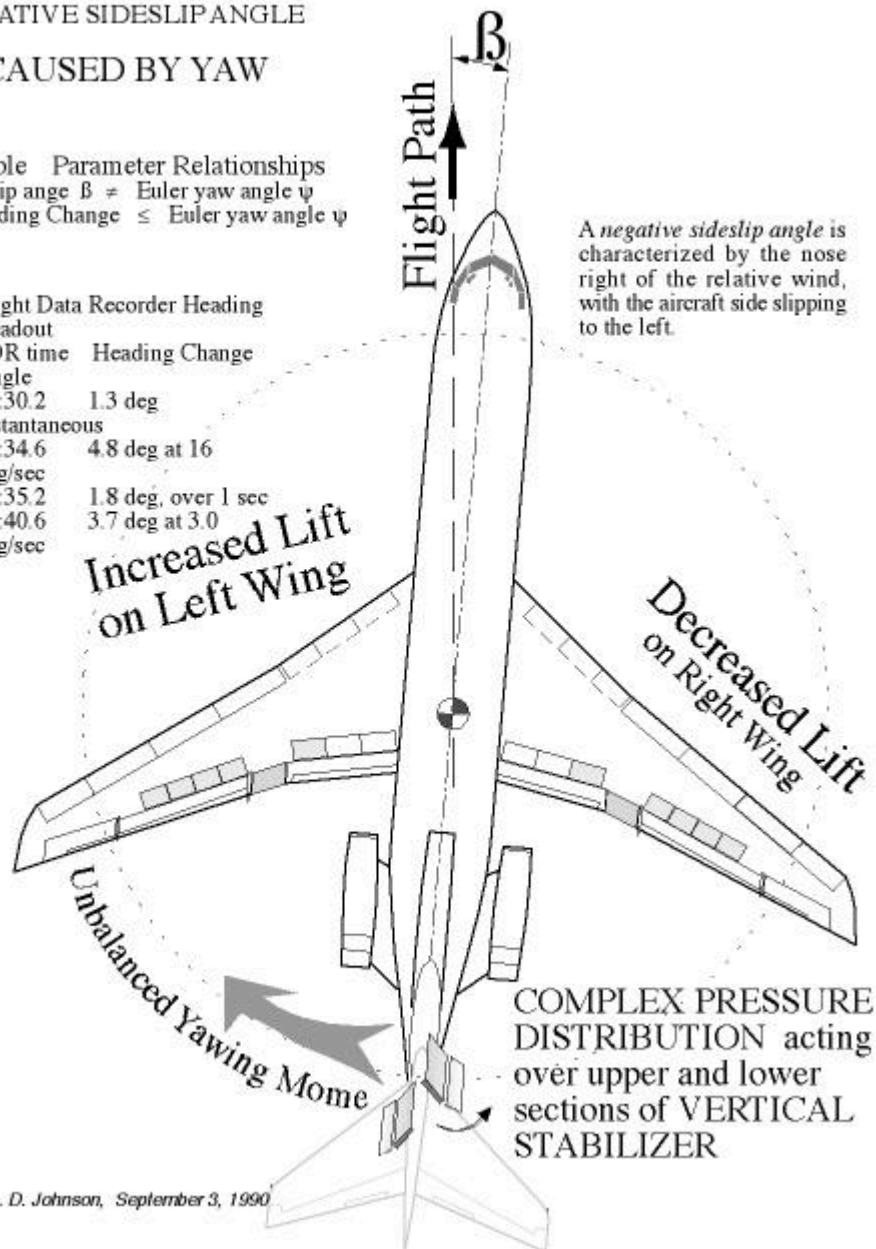
SIDESLIP

NEGATIVE SIDESLIP ANGLE

CAUSED BY YAW

Probable Parameter Relationships
 Sideslip angle $\beta \neq$ Euler yaw angle ψ
 FDR Heading Change $\leq$ Euler yaw angle ψ

Flight Data Recorder Heading
 Readout
 FDR time Heading Change
 Angle
 23:30.2 1.3 deg
 instantaneous
 23:34.6 4.8 deg at 16
 deg/sec
 23:35.2 1.8 deg, over 1 sec
 23:40.6 3.7 deg at 3.0
 deg/sec



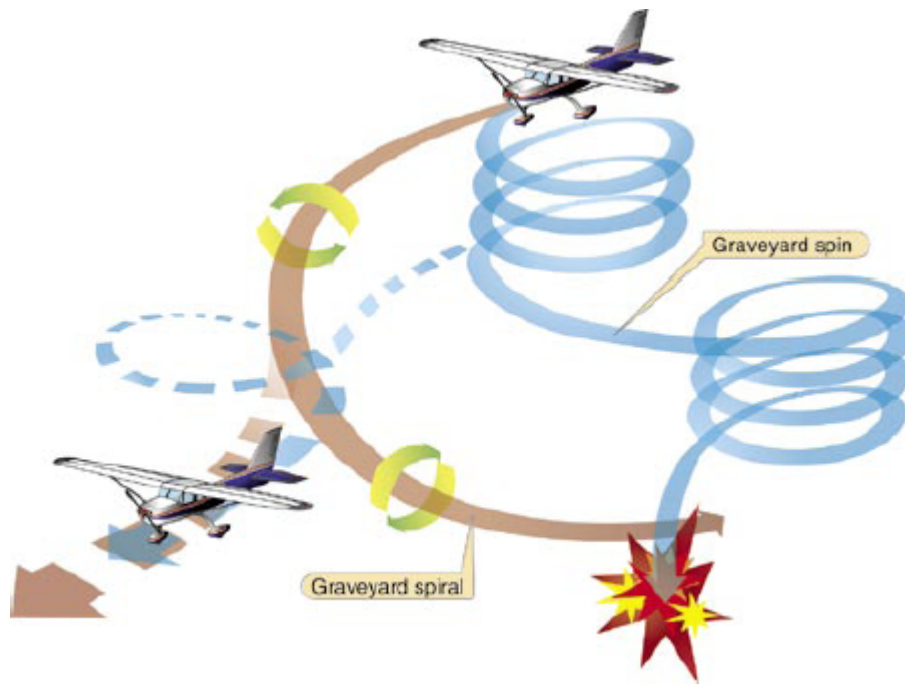
L. D. Johnson, September 3, 1990

Sideslip illustrated. Sideslip is created by the differences in the lift on the wings. It is a natural consequence of the flight regime and the design of the aircraft.

During a level turn, the yaw rate combined with the stabilizing yawing moment (due to yaw rate) results in a little bit of sideslip. The sideslip is positive for right turns, and negative for left turns.

That small amount of sideslip together with a stabilizing rolling moment due to sideslip (dihedral effect) results in the plane wanting to roll out of the turn. Which is exactly what we want.

With anhedral, the plane wants to roll *into* the turn, resulting in a “graveyard spiral”. This is something that we do not want. (Unless you are designing some kind of gravity directed munition...)

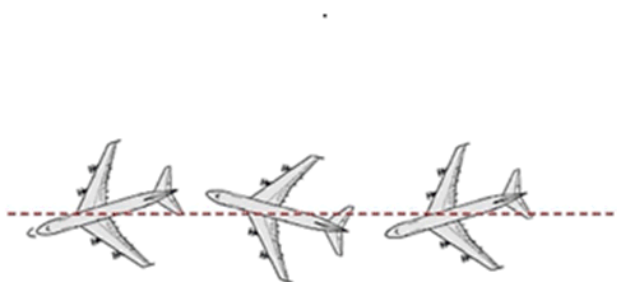


Graveyard spin compared to graveyard spiral. Image credit to FAA.

Now, to understand this tendency; the movement to roll into or out of a turn is called the “spiral mode”. This mode is controlled mainly using dihedral.

Most real airplanes have to limit the amount of dihedral they use to keep the Dutch roll mode (which is a rapid left and right oscillation), under control. While dihedral makes the spiral mode more stable, it reduces the damping of the Dutch roll.

TOP VIEW



FRONT VIEW



Here is a visualization of a dutch roll from both a bird’s eye view and a head-on view. The oscillation movement can clearly be seen.

You probably won’t have any Dutch roll problems with paper airplanes. This is likely due to increased yaw rate and roll rate damping that is associated with low airspeeds.

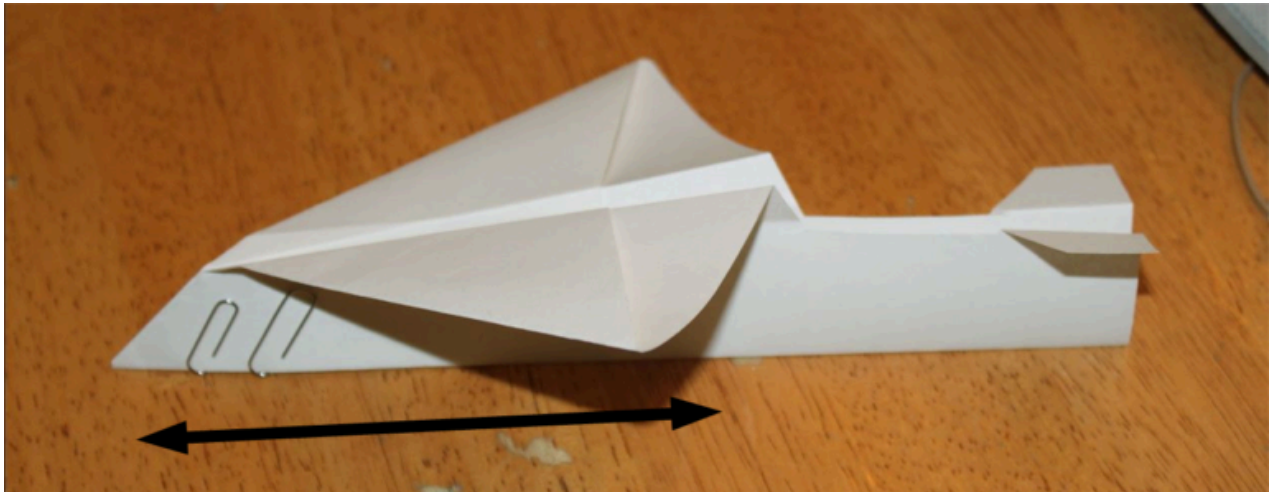
Because of this, all paper airplanes should be flown with plenty of dihedral.

| Paper airplanes should be designed with plenty of dihedral.

Up-Elevator

You should add a tiny amount of up elevator to the classic pointed nose paper airplanes. This will make sure they don't dive on you.

If you have the time and materials, you might want to add a few layers of tape or (maybe) a paper clip to the nose of the plane to improve its stability.



Place a paperclip on the airplane, and move it about to test the impact on the flight of the paper airplane.

The placement of a paperclip on a paper airplane can be used to move the center of gravity about on the plane. By adjusting the location of the paperclip, the stability of the plane can be improved.

Most “square” paper airplanes have plenty of weight in the nose, and require some up elevator to fly well. The amount of up elevator needed on a paper airplane is a pretty good indicator of its pitch stability.

So, when you build a paper airplane (of any kind) and place a paper clip on the nose. You need to go ahead and take it out for a few flights. Go ahead, make a few flights to determine the best amount of up elevator that is needed.

Now give it a few flights and see how it performs. Practice by moving the weight back slightly. Move it back a few cm, maybe a half an inch or so. Then try again. Keep on doing this. Experiment. See how the weight adds to the stability of the air plane.

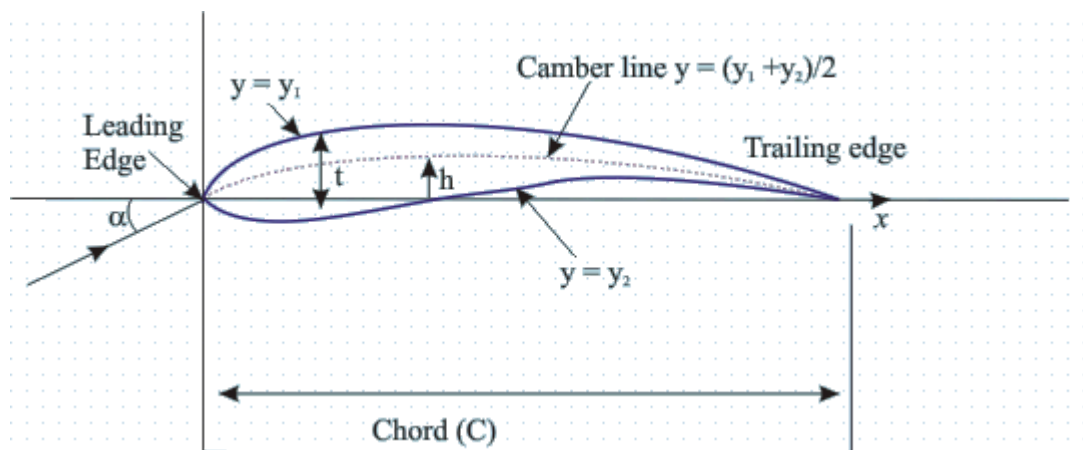
The amount of up elevator needed is reduced, and the plane becomes more sensitive to elevator adjustments. Now, take note. When the paper clip has been moved back to a point where you are using nearly no elevator deflection, and you can't get the plane to fly well, there is a reason. What is going on, is you have the CG at the neutral point (try to balance the plane on a finger, the point where it balances is the neutral point).

Airfoil shape

Contrary to what many people might think, airplane wings are not flat sheets of metal. They are curved. Which means that they are “Cambered”. Being cambered can be explained that if you took a cross section of the wing, it would look like a tear drop shape. In general, the top of the airfoil is rounded and the bottom is kept fairly flat.

That being said, paper airplane wings must be thin to work well.

Overall, the wings on the paper airplane need very little camber. Now, generally any curvature is limited to the front portion of the wing.



The cambered wing is designed to create areas of high and low pressure when air moves on the surface of the camber.

Generally, because most paper airplanes look a lot like “flying wings”, only small amounts of camber are practical. Now, large amounts of camber create nose down pitching moments which need tails to balance.

Hints and Tricks

Try to use a little curvature at the leading edge of the wing. You can use a pencil to round up the leading edges. You really do not need to have a fully cambered airfoil shape on a paper airplane. Typically, paper airplane performance is not noticeably degraded with flat, un-cambered airfoils. You know, the reason for this is likely due to low Reynolds numbers.

Reynolds number and Boundary layer thickness

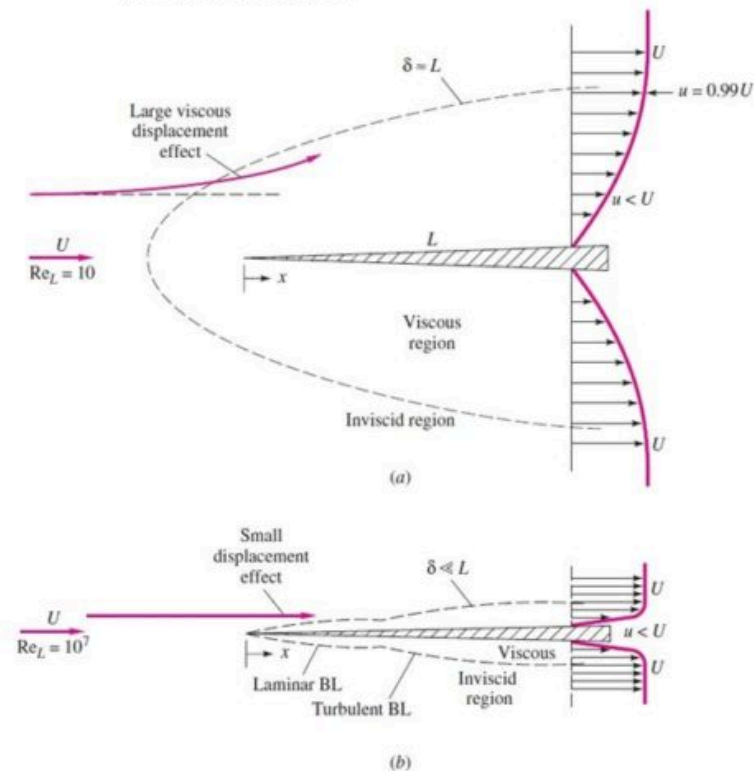


Fig. 7.1 Comparison of flow past a sharp flat plate at low and high Reynolds numbers: (a) laminar, low-Re flow; (b) high-Re flow.

Mecânica de Fluidos Ambiental 2015/2016

Movement of a fluid over a surface. The Reynolds number is the relationship between velocity, and the properties of the gas (or liquid) as a function of the shape of the surface.

To keep the paper airplane aloft for as long as possible, we need a turbulent boundary layer. Which is to say that the air is mixed up and confused; turbulent. We have to keep in mind that that a large portion of the boundary layer across the front of the wing is laminar flow, but for high lift we need a turbulent boundary layer.

Conclusion

I like to think that the most efficient way to learn is through play. As such, I agree with Fred Rogers. Play is the “work” for children. We learn how to become adults, socialize, think and build things through play.



The Syracuse University QUAD. It is the center of the university. As under graduates we would cross this area on our way to and from classes and the dorms.

While we were attending classes in aerospace engineering, it was our “hand’s on” play with paper aircraft that provided a visceral understanding of the basics of flight within the earth’s atmosphere. As such, I have never forgotten such adventures.

Of course, as young men, we were preoccupied in a mix of pursuits that all college boys engage in. Beer, girls, cigars and building paper airplanes (and shooting them down) ranked high on our enjoyable endeavors. I wanted to shoot them down with BB guns, but my liberal friends wouldn’t have any of that. However, they had no problems with fireworks, and bottle rockets. Combined, we would tend to get drunk, make experimental paper aircraft designs which we would launch, and then subsequently try to shoot down using manually targeted bottle rockets.

In the process we improved both our aviation design skills, as well as our understanding of basic rocket ballistics. We would chug down a beer as a reward for downing a paper airplane in flames. As I recall, we would also quaff down a beer if we had a close miss... Now, that I think about it, we also drank a beer if we missed by a long shot.

Maybe that is why we were always so drunk after trying to shoot down the airplanes...

Takeaways

- The building of paper airplanes can teach the basics of aviation design.
- Play is an efficient technique to learn.

- Those bottle rockets that lit up the sky decades ago over the Syracuse University campus were actually anti-paper airplane projectiles.
- Physical science isn't really all that complicated. People tend to use knowledge to obtain power and respect, instead of sharing for mutual benefit.
- Never discount the importance or value of fun.
- Don't be a "purist", paper airplanes can incorporate staples, paperclips, tape, glue, cardboard, and other features that could improve the design.

FAQ

Q: Why did you want to shoot down the paper airplanes?

A: We were drunk and at the time, it seemed like an appropriate thing to do. We started with some bottle rockets that one of our friends had. We became so enraptured with the concept that it became a semi-regular event. It was often done surreptitiously in the dark night. We would launch them from the windows in the day room of the dorm where all my friends lived.

Q: How did you get started in making paper airplanes at university?

A: My friend Peter was always fiddling with his hands. For a while, he began to start making paper airplanes. Over the months together, we started playing with them. We put them in races, and tried to find out how long that they would stay aloft.

Q: Did you ever get into trouble for launching bottle rockets?

A: Not me personally, but there were complaints. It was certainly against campus rules, but the rest of the dorm were either too stoned to complain, or too busy studying to notice. At that time, there weren't too many busybodies on campus. People had a Laissez-faire attitude about life. It was best summed up as "live and let live". So you could pretty much live your life on your terms, within limits of course. That is totally unlike today, where the campuses are populated with busybodies, and crybabies that are easily offended about everything.



American universities today are “safe spaces” where everyone can get offended by the slightest remark. It is Orwell gone amok. It’s a pretty bad situation. I, for one, am glad that I am not attending university today.

Q: Do you still make and fly paper airplanes?

A: No. Occasionally, I make one to two to amuse a child, but that is the extent of it.

Q: Did you ever make paper airplanes out of aluminum foil?

A: No, but it is an interesting concept. If the foil is thick enough it might lend itself to some interesting wing designs.

Posts Regarding Life and Contentment

Here are some other similar posts on this venue. If you enjoyed this post, you might like these posts as well. These posts tend to discuss growing up in America. Often, I like to compare my life in America with the society within communist China. As there are some really stark differences between the two.



The Most Important Things That a Man Looks for in a Woman.

Here, we look at the things that men search for in a woman. These things are considered, by men, to be absolutely essential in a mate. We discuss ten attributes and go into detail as to why they are important.



Adventures Exploring the Cemeteries of Indiana

Cemeteries can tell us a lot about ourselves, our history, and our future. This is most especially true in regards to smaller and remote cemeteries located in the counties that surround us. This is my story about the impressions that my forays into these curiosities made upon me.



What it was like being a kid in the early 1970's

This is my narrative of what it was like growing up as a young teenager in 1971. This took place in the hills of Pennsylvania outside of Pittsburgh. I was 13 years old. I went to school, rode my bicycle around everywhere. I fished with my father, and watched television a lot.



The Fiasco of the Reengineering of the Common Tomato

This is the sad, sad story about how the wonderful and tasty tomato was reengineered to be "better". It was made better to ship. Better to sell. Better to store. However, there was one thing missing. No one paid any attention to taste, which is why it tastes like a cardboard shipping box today.



A Movie Tribute to the 1980's When Ronald Reagan Ran

America. Here we argue that movies reflect society when they were made. Today we see popular movies being remade as all female versions due to the progressive realities today. Likewise, the movies of the 1980s reflected the time when they were made as well.



Mad Scientist Explorations

Here we discuss what it was like growing up as a boy in the 1960's and the 1970's and the people, things, and adventures that inspired us. As such, we explore how the independence of growing up enabled us to cope through life when we got older.



The Gorilla Cage in the Basement

When I was a small boy, I used to play in the abandoned buildings nearby. Here is a story about one of the objects that lay hidden inside one of those buildings and the possible mysteries that it held. As a child we thought it was a gorilla cage, now as an adult, I am not so sure...



The Importance of a Family Meal Together

There is nothing more important and satisfying than spending time together with friends and family and eating a meal together. It is a time of friendship, familial bonding and relaxation. Here we discuss what I have learned from my childhood and how I apply it today in my life.



The Pleasures of Fresh Baked Bread with Butter

One of the things that I have come to miss, since I moved to China, is fresh hard crusty rolls with full salted butter. The Chinese are not really fans of crusty bread. They like sweets and sponge cakes. Here I describe something that is close to me; not appreciating what you have right in front of you.



What Work was Like in the 1960's and 1970's.

When people go to work today, they find that there are all kinds of rules and regulations. It is for diversity, safety, inclusion, and improvement in a new progressive world. Well, here is what work was like before it became a sterile progressive testing ground for the New World Order.



Learning during my 1970s High School years

Here is a pretty long post on what it was like to go to elementary school in the 1960's, and then attend high school in the 1970's. I graduated in 1977, and that my friend is exactly the target date for the famous movie "Dazed and Confused". Here is the real deal what it was like.



Some True Stories of Cat Heaven

Being in MAJestic taught me many things. One of which was an intimate understanding of Heaven and other “spiritual” things relative to quanta entanglements. Here I discuss what it is like to have a beloved animal (cat) dies and what actually happens to them and why.



Life Lessons from Working within the Corporate Dream – 1

After I was trained by MAJestic, I was left to fend for myself in Corporate America. It was a hellish existence. As such, I learned many lessons concerning behavior and the ultimate impact that it holds on our families, our friends, our loved ones, and our very lives. Here are my commentaries.



Life Lessons from Working within the Corporate Dream – 2

This is part two of the two part section. There is quite a lot of information to impart. Stories are told, and experiences relayed. Honestly working in corporate America in the 1980's, through the new century really sucked. It was harsh on everyone except the owners of the companies.



How to Build Up Your Life from Nothing.

This is for all those young men who are just leaving school and trying hard to build up their life. You can do it. Here are my thoughts and advice for a young man who is encountering that most difficult of challenges; carving your life out in the wilds of the world.



Allow your Children to Play and Grow (Part 1)

I am horrified, just horrified, that parents do not allow their children to play, roam and explore on their own. It has created a society of pampered fearful children. Here I discuss this in terms of what I know. This is very non-PC and might offend the more liberal readers out there.



Allow your Children to Play and Grow (Part 2)

This is the second part of my rant about permitting children to play, explore and have fun. Everyone, the odds that some stranger will abduct your child is very small. Most people who do this are family friends and neighbors. Stop being so fearful. It harms the development of your child.



Ode to Diabolical Cretin John McCain

Well, that evil son-of-a-bitch is finally dead. If his grave wasn't so far away, I go there and dance on top of it. This guy has spent his life backstabbing people, hurting people, concocting wars, and doing everything in his power to disrupt the lives of people. Good fucking riddance.



The Song "Baby Got Back" Translated into Latin

Here is a fun post. The song "Baby Got Back" is translated into Latin and then back again. I have to tell you that it is indeed hilarious. *Rebecca, ecce! tantae clunes isti sunt!* (Rebecca, behold! Such large buttocks she has!) Anyways, it's just all fun and games. Enjoy.

More Posts about Life

I have broken apart some other posts. They can best be classified about ones actions as they contribute to happiness and life. They are a *little different*, in subtle ways.



On Being an Older Gent

Here are my thoughts on growing old as a man. The truth is that the life that I expected to have as I was growing up never materialized. Instead, something else manifested. My life today is nothing compared to what I thought it would be. For my illusions were created by American media.



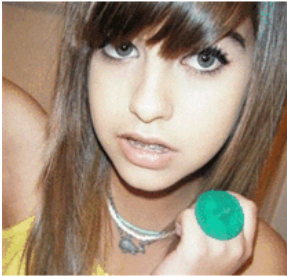
Calexit and the American Civil War

This is a pretty detailed look at the causes of the American civil war, and why it wasn't about slavery. We look at the rewriting of history by the internet, and the efforts that President Obama took to keep California intact so that the Democrats could maintain political power and control.



Why an American man should leave America and Travel

All people should get up off their duff and go forth and Travel. However, this is most especially true for American men. The United States has become a crucible of PC progressive realities that attack and depreciate American men. Go out, young man, see the world as it really is!



The Amazing Bremelanotide PT-141

Forget Viagra, and all those other ED medications. This is the real deal. This little baby causes humans to *go into heat*. It interacts with the brain and sets up feelings of intense arousal on both men and women. It turns people into 16 year olds. As such it is banned in the USA by the FDA.



Bronco Billy and the 25th Law of Power

Bronco Billy is a fictional movie starring Clint Eastwood. The movie depicts the life of a man who decides to change his life and go ahead and do what he has always dreamed of doing. He goes out and becomes his dream. There are lessons here for all of us, we only need listen.



The r/K Selection Theory applied to American Politics

The r/K theory is a pretty well-known theory on animal behavior. It simply states that animals behave as they need to based on the scarcity of resources. Here, we argue that it can be applied to why humans follow different political philosophies. Especially in American politics.



How they get away with it

Ten years ago, Bankers committed fraud in Wall Street that just about devastated the nation. Earlier, we had the S&L fiasco, the nightmare with penny stocks, and now the grand wholesale treason with the selling of Top Secrets to a foreign nation. This is how they get away with it.



The Line in the Sand – Now What?

It's the same story. Just vote for us, and we will correct everything. Yet, nothing ever happens. You know, boys and girls, this old "song and dance" can only be maintained for so long. Eventually it will fracture, and people will no longer rely on elections to institute change...



The importance of having a second passport.

So President Trump is building a wall to protect Americans. Hum. Where have I heard that before? Oh yes, in Germany right when they were making the wall that separated Berlin. Walls serve two purposes; they keep people out, but they also keep people in. Be careful what you wish for.



How Rocket Scientists Build Paper Airplanes

This is a tale of what me and my classmates used to do when we were young Aerospace Engineering students at the university. We would design paper airplanes and shoot them down with bottle rockets. Here is just a fun little exercise down "memory lane". Enjoy.



What is Snopes.com and can it be Trusted?

Snopes is a "fact check" website that is fully funded by the oligarchy. It is used to "shout down" any narratives that does not support the rich and powerful, and politically connected globally. Here we discuss the organization, who they are and what they represent.



Today America Taxes it's Expats without Representation.

This is a sad state of affairs. Americans who reside outside of the United States now have to pay American taxes, even though they have no representation in Congress. As I recall this was one of the reasons for the American cry for independence in 1776.



What life is like inside the ADC Prison in Arkansas

This post discusses what life is like in a hard labor prison in Southern Arkansas. We discuss hoe squads, food, the hole, commissary, dress, fashion, homosexual culture, prison gangs, murders, relationships and other aspects of life when you are sentenced to “Hard Time”.



The Hazing of New Employees – A Lost Tradition

Up until the 1980's all companies would haze new employees. Today, we live in a world regulated by the government and policed by Human Resources. We forget how things used to be just a handful of decades ago. Here is my narrative when I started to work.



How to Build a Gallows and a Hangman's Noose

The United States is entering a period of strife. There will be violence. There will be discord and discomfort. It will be an ugly time. There will be winners and losers. It will be painful for the losers, and the victor will need build a gallows. This is how to do so and why it is needed.



The Ultimate Method to Make Your Computer Safe and Secure

The best way to make your computer secure and private is to have very strong encryption and use an obscure operating system that very few people know about. Here we look at 37 obscure computer operating systems.



Is America a Republic or is it a Democracy – What is it?

Here we take a nice good look at what the United States is today. We look at reports that are suggestive of other governmental organizations. We look at what is going on contemporaneously inside the United States, and we look at how the United States behaves globally.



The Many Things that Democrats want to Ban

This is a running list of the many things that American Democrats want to ban, have banned, or are trying to ban. You know, a ban of anything is a restriction on freedom. That is not what America was founded upon. Yet, here we are today. Everyday is a new ban, and loss of freedom.



Nations that an American Man could Travel and Relocate to – Part 1

Here, I suggest places where an American man could go if he wanted to start all over again. You know, this happens. Often a man will go through a nasty divorce, or get raked over the coals by a business associate or the United States government. When it happens, it's time to start new.



Nations that an American Man could Travel and Relocate to – Part 2

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Stories that Inspired Me

Here are reprints in full text of stories that inspired me, but that are nearly impossible to find in China. I place them here as sort of a personal library that I can use for inspiration. The reader is welcome to come and enjoy a read or two as well.



Robert Heinlein's "The Long Watch" – Full Text

What if you were in charge of all the nuclear weapons in the world, and your base Commander checks your political leanings to see if you are willing to conduct a coup d'état. What would you do? Would you go along with the opportunity, or would you be a lone hero?



Kaleidoscope (Full Text) A Story by Ray Bradbury

This is the full text of a short story by Ray Bradbury. It concerns the final moments of a crew of spacemen as they drift about in space helplessly after their space rocket explodes. They know that no one can help them as they await their final moments tumbling through space...



Here There Be Tygers by Ray Bradberry

When I was younger I used to read science fiction stories for pleasure. They filled my minds with adventure and passion. One of my favorite authors was Ray Bradberry and here is a complete reprint of one of the stories that has influenced me to be the man that I am today. Enjoy.



The Long Rain (Full Text) by Ray Bradberry

This is the full text of a most excellent science fiction story. It concerns a group of survivors from a crashed spaceship stranded on a Hellish rainy planet; the tropical cloud-covered Venus. They survive the crash, but have to battle the forces of nature and their own passions.



The Fog Horn (Full Text) by Ray Bradberry

Here is a story about the emotions of being the last of your kind. This story is one that influenced me when I was a young boy in the 1970's and continues to haunt me to this day. All of us are human. We all feel emotions and passions. They drive us to accomplish or destroy. It's our choice.



The Rocket (Full Text) A Story by Ray Bradberry

This is a nice short story about a poor man who dreamed about flying into space. No matter how hard he worked, and no matter how much he saved, he was never able to provide for that dream. Then one day, when he was asked to scrap a test rocket, a thought came to mind...



Dark They Were and Golden Eyed (Full Text) by Ray Bradberry

This is the complete short story by Ray Bradberry. It is a story about change and adaptation. I first read this story in my teens and it influenced me to live the life that I am currently living. It is a great read.



A Sound of Thunder (Full Text) A Story by Ray Bradbury

This is the full text of the great time-travel story by Ray Bradbury. Here, a time-travel machine is used to send people back in time to shoot a dinosaur. However, something goes wrong. What would you do? It's a classic story and well worth a read again.



The April Witch (Full Text) by Ray Bradbury

This is a story written by Ray Bradbury that describes what might happen when a young teenaged girl has the power to enter the body of another person. What if, forbidden to experience love and relationships with "normal" people, she uses magic to experience forbidden pleasures...

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Notes

1. Compiled 19JUL18.
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